

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Thursday, 20 August 2026, 8:44 AM**State** Finished**Completed on** Thursday, 20 August 2026, 8:45 AM**Time taken** 1 min 17 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include <stdio.h>
2 | int main() {
3 |     int num, count = 0;
4 |     if (scanf("%d", &num) != 1) return 0;
5 |     for (int i = 1; i <= num; ++i) {
6 |         count++;
7 |         count++;
8 |         if (num % i == 0) {
9 |             count++;
10 |        }
11 |    }
12 |    count++;
13 |    printf("%d", count);
14 |    return 0;
15 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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